ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

COLLEGE OF ENGINEERING

INTEGRATED ENGINEERING TEAM PROJECT

A SYSTEM DESIGNED TO CAPTURE AND REMOVE CO_2 FROM THE AIR

Group 36

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ABSTRACT

This project presents the design of a system aimed at capturing and removing CO_2 from the air, addressing the growing concerns of air pollution and its environmental impact. The system utilizes a CO_2 absorption method involving a chemical reaction with sodium hydroxide (NaOH) solution, which absorbs CO_2 from the surrounding environment. The project aims to provide an innovative solution to mitigate the increasing levels of CO_2 in the atmosphere, contributing to environmental conservation efforts.

The design process involved detailed engineering calculations, including airflow analysis, material selection, power requirements, and CO_2 absorption efficiency. The system's key components include a PVC chamber for the NaOH solution, a pump to regulate airflow, and a sensor to measure CO_2 levels in the outlet air. The system is designed to operate with minimal power consumption and maximum CO_2 absorption capacity, ensuring both efficiency and sustainability.

By calculating the airflow rate, the capacity of the NaOH solution to absorb CO_2, and the power requirements of the system, we have developed a prototype that can effectively capture and process CO_2. The system's efficiency is further validated through a comprehensive analysis of its absorption rate and overall performance, ensuring that it meets the project's goals of reducing CO_2 emissions and contributing to a cleaner, healthier environment.

This project serves as a step forward in developing practical, cost-effective solutions to combat climate change and air pollution, with potential applications in various industries and urban environments.

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CHAPTER ONE

1. INTRODUCTION

1.1. Project Overview

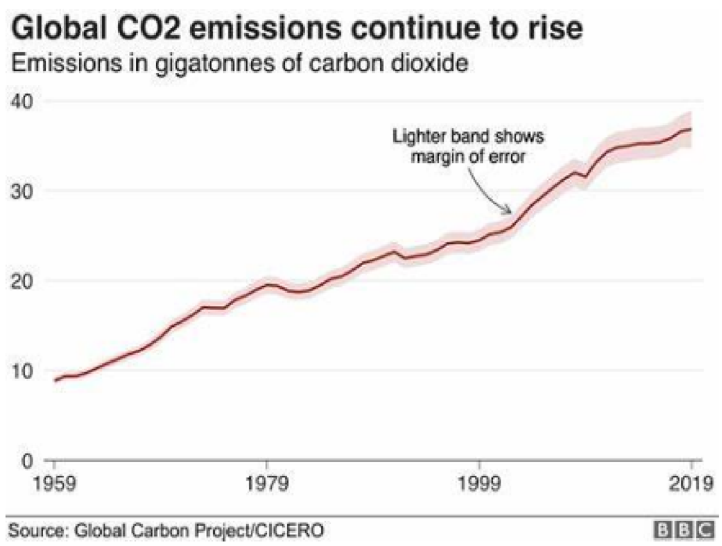
Our project focuses on designing an innovative air purification system to reduce atmospheric CO_2 levels, aiming to improve air quality, support public health, and contribute to climate action. Using a chemical absorption chamber with sodium hydroxide (NaOH) solution, this system captures CO_2 from the air, converting it into sodium bicarbonate, which effectively removes CO_2—a major greenhouse gas. Equipped with a gas sensor, the system assesses Co2 level to give real-time display and monitoring

While CO_2 emissions are a global challenge, this project is designed with a broader global application in mind, addressing air quality in urban environments. The system holds potential for deployment in cities worldwide to mitigate air pollution and align with sustainable development goals.

By focusing on reducing urban CO_2 concentrations, this project promotes healthier and more sustainable spaces, directly addressing climate action and public health concerns. Its scalability ensures relevance for global cities striving for environmental sustainability

1.2. Background of the Project

Carbon dioxide is a trace gas that plays an integral part in the greenhouse effect, carbon cycle, photosynthesis, and ocean carbon cycle. It is one of the three main greenhouse gases in the atmosphere of Earth. The concentration of carbon dioxide in the atmosphere reached 0.04% in 2024, The growing levels of atmospheric carbon dioxide have become a significant environmental challenge in recent years, due to the fact that CO2 is one of the greenhouse gases responsible for causing global warming. This is a graph depicting the growth of the co2 production after the Industrial Revolution and introduction of vehicles



The increase in atmospheric CO2 concentration is largely caused by the combustion of fossil fuels for energy generation, deforestation, and agricultural practices. To safeguard the planet for future generations, it is imperative to implement measures to decrease CO2 emissions into the atmosphere. This is a bar chart describing the sources and destinations of CO2 that have been recorded since 1850.

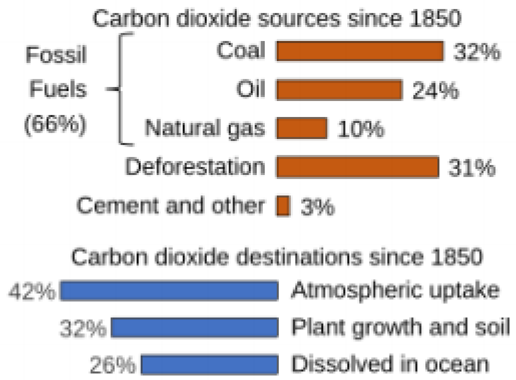


*Figure STYLEREF 1 \s 1. SEQ Figure * ARABIC \s 1 2 Sources of Co_2*

As shown in the chart the uptake of CO2 into the atmosphere is the highest among the three, so we decided on a project that aims to reduce the CO2 levels in the immediate surroundings escalating the concentration of carbon dioxide (CO2) in the Earth's atmosphere is a primary driver of climate change, leading to significant global environmental and socio-economic challenges. The burning of fossil fuels, deforestation, and industrial processes have resulted in an unprecedented build-up of atmospheric CO2. This led us to search for effective carbon capture technologies. Chemical absorption, which involves the reaction of CO2 with a suitable solvent, has caught our attention. Among the potential solvents, sodium hydroxide (NaOH) is considered a promising candidate due to several key advantages. NaOH is readily available, relatively inexpensive, and exhibits a high capacity for CO2 absorption. The reaction of CO2 with NaOH results in the formation of sodium carbonate (Na2CO3), effectively removing CO2 from a gas stream

1.3. Statement of the Problem

The rising levels of carbon dioxide (CO_2) in the atmosphere, driven by activities like burning fossil fuels and deforestation, are major contributors to climate change. With CO_2 concentrations at an all-time high, it's crucial to find ways to reduce its impact. This project focuses on developing a practical solution to capture and lower CO_2 levels in the air. By using sodium hydroxide ($NaOH$) to absorb CO_2, we aim to create an effective system that helps mitigate the environmental effects of rising CO_2 and combats global warming.

1.4. Objectives

1.4.1. General Objectives

The general objective of this project is to design, develop, and create a workable prototype of a CO_2 capture system using sodium hydroxide ($NaOH$) to reduce the concentration of carbon dioxide in the surrounding environment and contribute to mitigating the effects of climate change.

1.4.2. Specific Objectives

The specific objectives of this project are:

1. To design a body structure suitable for installing the CO_2 capture system in various environments and locations.
2. To design a mechanism for capturing and removing CO_2 from the surrounding environment effectively.
3. To develop an integrated system for controlling CO_2 reduction levels using sensors and a mobile app for real-time monitoring and adjustments.
4. To design a product circuit system for the CO_2 capture system installation using Proteus for simulation and prototyping.
5. To program the sensors to accurately detect and measure the CO_2 levels in both the intake and outlet air streams, ensuring proper system functionality.
6. To implement a feedback system that adjusts the CO_2 capture mechanism based on real-time sensor readings and data from the mobile app.
7. To optimize the system for practical use, ensuring ease of installation, maintenance, and scalability in various environments.

1.5.Scope of the project

The scope of this project focuses on the design, development, and creation of a functional CO_2 absorption unit, along with integrating sensors for real-time monitoring and data collection. The key areas of focus are:

1. **Design and Development:** The project will involve the conceptualization and design of a CO_2 capture system using sodium hydroxide (NaOH) for CO_2 absorption. This includes the development of a **workable prototype** that demonstrates the system's functionality in reducing CO_2 levels.
2. **Sensor Integration:** The system will integrate sensors to continuously monitor CO_2 concentrations in real-time, ensuring accurate measurements and enabling effective control of the CO_2 reduction process.
3. **Simulation:** The system's design will be simulated using tools like Proteus to test and optimize the system's components and functionality before proceeding with the physical prototype.

1.6. Limitations of project

While the project aims to deliver a functional prototype and system design, several limitations were encountered during its execution:

1. **Coordination Challenges:** The project team consists of 11 members from different fields, which made it difficult to coordinate and integrate the various components of the project effectively. The diversity of expertise required more time for collaboration and alignment of ideas.
2. **Material Shortages:** There was a shortage of materials needed to manufacture the prototype, which slowed down the progress of the project and impacted the quality of the initial prototype.
3. **Lack of Support:** The team faced challenges due to a lack of support in terms of resources and guidance. This made certain tasks, such as troubleshooting and refining the prototype, more difficult.
4. **Limited Budget:** Due to the project's constrained budget, it was not possible to acquire all the required materials and equipment. However, the project advisor helped

by providing some resources from his own collection, which allowed the team to continue the work.

Despite these limitations, the project is still in its early stages, with the initial session focused on building the prototype. The project is open to further analysis and optimization to enhance its performance and functionality in the future

1.7. Project Management

This project aims to design and develop a CO_2 absorption system that aligns with Sustainable Development Goals (SDGs), particularly SDG 13 (Climate Action) and SDG 2(Good Health and Sanitation). By reducing atmospheric CO_2 levels through an integrated mechanical and chemical approach, this system contributes to mitigating climate change while fostering interdisciplinary collaboration.

1.7.1. Project Phases and Timeline

The project is structured into distinct phases, each aligned with the timeline and activities outlined in the Integrated Engineering Team Project (IETP) framework:

1. **Initiation (Weeks 0-1)**
    - Team formation and advisor assignment.
    - Orientation and project guideline release.
    - Activity: Discuss project scope, objectives, and initial ideas with the advisor.
2. **Planning (Weeks 2-4)**
    - Submit project ideas for approval.
    - Develop detailed design, requirements, and project plan using Form A.
    - Conduct training sessions in the mechanical workshop for model construction.
3. **Execution (Weeks 5-9)**
    - Prototype construction using NaOH solution, air pump, and fans.
    - Integration of Arduino and sensors to track CO_2 levels.
    - Activity: Data collection and performance testing to refine the design.
4. **Optimization and Testing (Weeks 10-12)**
    - Optimize system components for maximum efficiency.
    - Perform extensive testing under different environmental conditions.
5. **Delivery (Weeks 13)**

    - Prepare a functional prototype demonstration.
    - Submit a comprehensive report and a 3-minute project video.

1.7.2. Resource Allocation

- **Team Roles:** Each team member contributes based on their discipline, such as chemical engineering for absorption processes and software engineering for app development.
- **Facilities and Tools:** Mechanical workshop, Arduino kits, CO_2 sensors, and chemical materials.
- **Supervision:**
  Regular advisor meetings for progress reviews and feedback.

1.7.3. Milestones

- **Week 2:** Project idea submission and approval.
- **Week 4:** Project Proposal submission.
- **Week 5:** Start of prototype construction.
- **Week 11:** Completion of initial optimization and testing.
- **Week 12:** Prototype ready for final demonstration.

1.7.4. Risk Management

Potential risks include delays in material procurement, technical challenges in system integration, and inefficiencies in CO_2 absorption. Mitigation strategies include maintaining buffer periods in the timeline, conducting preliminary testing, and consulting with advisors for troubleshooting.

1.8. Alignment with SDGs

This project directly supports SDG 13 (Climate Action) by providing an innovative solution to reduce CO_2 emissions. The interdisciplinary approach also aligns with SDG 9 (Industry, Innovation, and Infrastructure) by fostering technological advancements and sustainability.

CHAPTER TWO

2. LITERATURE REVIEW

Carbon dioxide (CO_2) emissions significantly contribute to climate change, prompting the development of innovative capture methods. Sodium hydroxide (NaOH) offers a promising solution through chemical reactions with CO_2 to form sodium carbonate (Na_2CO_3) or sodium bicarbonate ($NaHCO_3$). These reactions, expressed as $2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$ and $NaOH + CO_2 \rightarrow NaHCO_3$, are reversible, allowing CO_2 release and NaOH regeneration, making the process sustainable [1], [2].

Key factors influencing efficiency include NaOH concentration, temperature, and pH control. Higher NaOH concentrations enhance absorption but may cause side reactions, while lower temperatures favor bicarbonate formation, reducing energy demands for regeneration. Optimal pH minimizes side reactions and ensures efficient absorption [3], [4]. Despite its effectiveness, the energy-intensive regeneration process, requiring temperatures over 600°C, presents challenges. Renewable energy integration, such as solar or geothermal, could mitigate these issues [5].

NaOH-based systems are environmentally appealing, offering cost-effectiveness and CO_2 reuse in industrial applications like sodium bicarbonate production. However, their sustainability relies on improving energy efficiency and integrating renewable sources [6]. Recent innovations, such as electrochemical regeneration and hybrid systems combining NaOH with other technologies, show promise in reducing energy costs and improving scalability [7].

NaOH-based CO_2 capture systems are effective but face challenges in energy demands and scalability. Future efforts should focus on optimizing regeneration, integrating renewable energy, and exploring hybrid solutions to advance this promising technology [8].

CHAPTER THREE

3. THEORY, METHODOLOGY, AND PROCEDURE

3.1. Theory

The CO_2 capture system operates on the principle of chemical absorption, using sodium hydroxide (NaOH) to remove carbon dioxide (CO_2) from the air. The reaction is highly effective, as CO_2 reacts with NaOH to form sodium carbonate (Na_2CO_3) and water, and further reacts to form sodium bicarbonate ($NaHCO_3$):

$$CO2+2NaOH \rightarrow Na2CO3+H2O$$

$$Na2CO3+CO2+H2O \rightarrow 2NaHCO3$$

This chemical process is efficient even at low atmospheric CO_2 concentrations, such as 400 ppm. The absorption efficiency depends on factors like the NaOH concentration, controlled airflow, and solution saturation levels. As NaOH reacts with CO_2, the solution's capacity to absorb decreases, necessitating replacement or regeneration to maintain functionality.

Fluid dynamics principles are applied to optimize the reaction. The airflow rate, calculated as:

$$Q=A \cdot v$$

- where Q is the airflow rate
- A is the cross-sectional area
- v is air velocity

To ensure proper exposure of CO_2 to the NaOH solution. Fans and air pumps are used to maintain consistent airflow, enhancing the system's absorption capacity.

The system is designed to be modular and scalable, making it suitable for small applications, such as homes, or large-scale industrial setups. Safety protocols are integral to the design, addressing the caustic nature of NaOH and ensuring proper handling, storage, and disposal of byproducts.

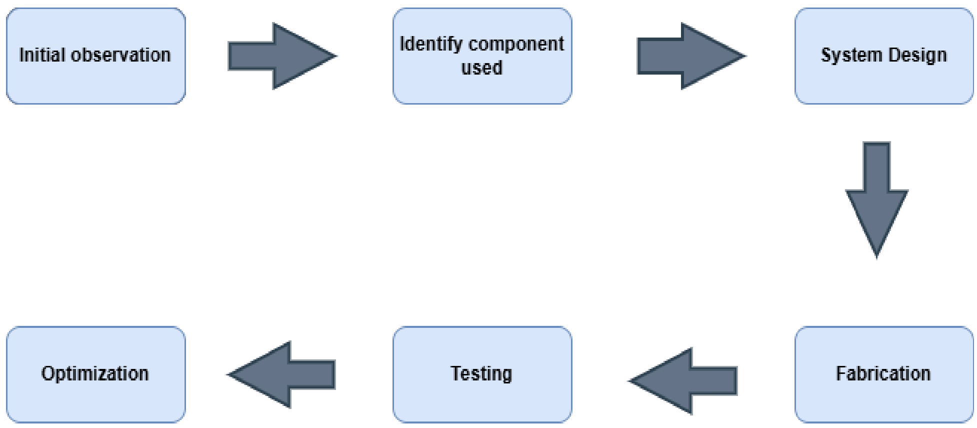
Environmentally, the system aligns with sustainability efforts by reducing atmospheric CO_2, improving air quality, and mitigating climate change impacts. By leveraging proven chemical and engineering principles, the system offers an effective and scalable solution to reduce carbon emissions.

3.2. Methodologies

The procedure for designing and developing the CO_2 absorption system involves hardware development, software design, sensor integration, and system testing. The system consists of an air pump, fans, NaOH solution, a gas sensor, an Arduino microcontroller, and a Flutter app for real-time data tracking. The Arduino controls the system components based on CO_2 levels detected by the sensor.

The project starts with assembling the hardware, including the Arduino, gas sensor, fans, and pump. The gas sensor is calibrated to detect CO_2 levels at the inlet and outlet. The Arduino is programmed to adjust the system based on these readings, and data is sent to the Flutter app for monitoring.

A physical prototype is built to test the system's functionality, with multiple iterations of testing to refine the system's performance. Data from literature and online resources guide the development and troubleshooting process.



3.3. Synthesis

The development of the CO_2 absorption system began with a review of existing literature and identification of the key components required to meet the system's objectives. The procedures involved are outlined below, starting with problem formulation and solution strategy. We used various resources, including textbooks, academic papers, and online references, to guide the system design. The method for problem-solving was visualized using software tools like Visio, helping to conceptualize the system's operation and integration.

3.4. Data Collection and Analysis

For effective project completion within the time constraints, secondary data collection was employed. Sources like textbooks, journals, online resources, and research studies were consulted. This approach helped reduce error, save time, and provide a broad range of qualitative and quantitative data. The collected data was then analyzed to meet the specific objectives of the CO_2 absorption system. Quantitative data such as material types, dimensions, cost, and scale were selected according to the project's specifications, ensuring the design would meet operational requirements.

3.5. Material Selection

The bill of materials (BOM) for the CO_2 absorption system includes the following components:

- Arduino Uno
- MQ-135 CO_2 gas sensor
- Air pump
- Fans
- NaOH solution
- Jumper wires
- Breadboard
- DC power supply
- Plastic box and sheet

1. MQ-135

The MQ-135 gas sensor is used to detect CO_2 and other gases like ammonia, alcohol, and benzene. It works based on a metal oxide semiconductor that changes its resistance in response to the presence of gases. The sensor provides an analog output, which is read by the Arduino to measure CO_2 concentrations. When the CO_2 level exceeds a predefined threshold, the Arduino adjusts the operation of the air pump and fans to regulate air quality. This sensor was chosen for its sensitivity, affordability, and ease of integration into the system.

Figure 3.2 MQ-135 gas sensor

2. Air Pump and Fans

The air pump and fans are used to circulate air through the NaOH solution, facilitating CO_2 absorption. The pump draws air into the system, while the fans assist in creating airflow through the NaOH solution, where the CO_2 is absorbed and neutralized. These components were selected for their efficiency in air circulation and their compatibility with the Arduino control system.

3. Arduino Uno

The Arduino Uno microcontroller acts as the central control unit of the CO_2 absorption system. It processes inputs from the CO_2 gas sensor and controls the air pump and fans based on sensor readings. The Arduino Uno was chosen for its ease of use, open-source nature, and compatibility with various sensors and components. It is powered via USB or an external power supply, making it flexible for different setup configurations.

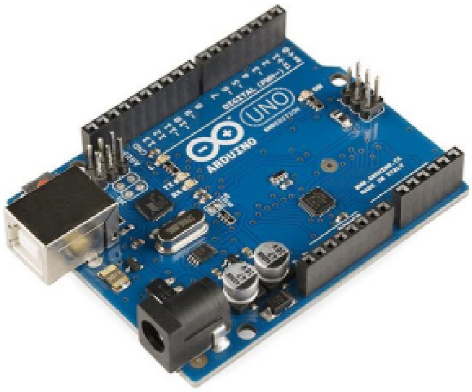


Figure 3.3 Adruion UNO

3.6. System Design

3.6.1. System Design Overview

The **CO_2 Absorption System** consists of two primary functions: **air intake** and **air exhaust**, both of which are integrated into a closed-loop system for efficient CO_2 capture. The system works as follows:

1. **Inlet (Air Intake)**:
    - **MQ-135 Gas Sensor**: The MQ-135 gas sensor is positioned at the **air intake** to detect the concentration of CO_2 in the incoming air. It measures the levels of CO_2 and other gases, providing real-time data about the air quality.
2. **NaOH Absorption Chamber**:
    - Once the air enters the system, it is directed through a **NaOH solution**. The sodium hydroxide (NaOH) reacts with the CO_2 in the air, capturing it and neutralizing it. The chemical reaction transforms CO_2 into sodium carbonate and sodium bicarbonate, effectively removing it from the air.
3. **Fans and Air Pump**:
    - The **air pump** draws air into the system, while **fans** create a consistent airflow through the NaOH solution. This ensures that the air stays in contact with the solution long enough for maximum CO_2 absorption.
4. **Exhaust (Air Outflow)**:
    - After passing through the NaOH solution, the CO_2-depleted air is expelled from the system through the **outlet**, where the concentration of CO_2 is significantly lower.
5. **Arduino Uno**:

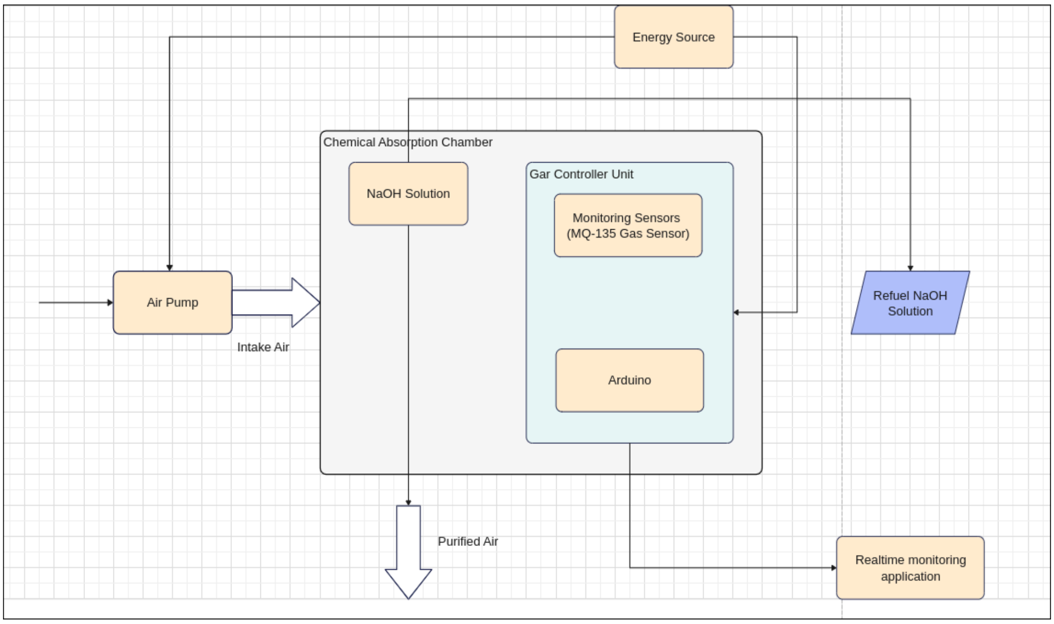
- The **Arduino Uno** acts as the control unit. It continuously receives data from the MQ-135 gas sensor, which monitors the CO_2 concentration in the inlet air. Based on the sensor data, the Arduino adjusts the operation of the air pump and fans to regulate airflow and optimize the CO_2 absorption process.

6. **Real-Time Monitoring**:
   - The Arduino is connected to an **app** that provides **real-time data** on the CO_2 concentration levels, system status, and other metrics. The sensor data is continuously sent to the app, allowing users to monitor the system's performance remotely.

3.6.2. Procedure

1. **Air Intake and CO_2 Detection**:
   - Air enters the system through the inlet, where the MQ-135 gas sensor measures the CO_2 concentration.
2. **CO_2 Absorption Process**:
   - The air passes through the NaOH solution, where the CO_2 reacts with NaOH and is removed from the air.
3. **Control Mechanism**:
   - The Arduino monitors the CO_2 levels from the sensor. If the levels exceed a predefined threshold, the Arduino adjusts the pump and fan operation to maintain optimal airflow and ensure efficient CO_2 capture.
4. **Real-Time Data Monitoring**:
   - The Arduino sends real-time sensor data to the app, where users can track the CO_2 concentration levels and monitor system performance.
5. **Air Outflow**:
   - The air exits the system, now depleted of significant amounts of CO_2, through the exhaust.

This design allows for continuous monitoring of air quality and provides an automated mechanism for regulating CO_2 capture in real time, ensuring efficient operation.



As shown in Figure 3.4, the process begins with air being sucked into the air pump, which directs it into the CO_2 absorption chamber. Inside the chamber, CO_2 is absorbed by the sodium hydroxide solution, effectively reducing the concentration of harmful emissions. The purified air is then released back into the environment, completing the cycle. The sensor continuously measures the Co2 level to monitor the effectiveness of the system. The design ensures minimal energy consumption while maintaining maximum CO_2 absorption efficiency

Conceptual Designs Considered

The design of this CO2 purifier prioritizes simplicity, efficiency, and scalability. Using readily available materials like PVC pipes and fittings, the design is straightforward to assemble. The strategic placement of fans and the use of an air pump maximize CO2 absorption efficiency by ensuring optimal airflow and circulation. Additionally, the design's modular nature allows for easy scaling up or down to accommodate different needs and spaces.

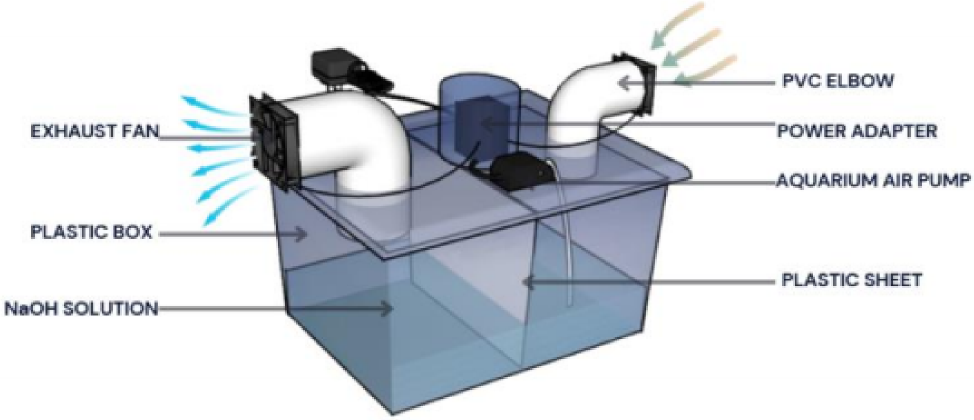


Figure 3.5 Design for prototype

The design for an industrial CO2 purifier is a scaled-up version of the prototype, optimized for large-scale applications. It features a sturdy metal enclosure with a powerful industrial-grade fan to ensure efficient air circulation and CO2 removal. The unit is equipped with multiple power outlets for easy integration into industrial settings. This design offers a robust and reliable solution for purifying air in industrial environments with high CO2 concentrations.

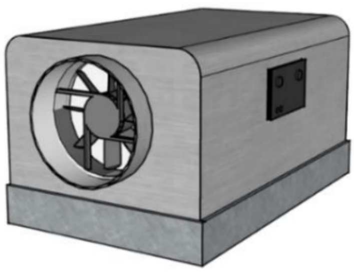


Figure 3.6 Design for an industrial setting

CHAPTER FOUR

4. DEVELOPMENT, ENGINEERING AND ECONOMIC ANALYSIS

4.1. Implementation Steps

1. **Air Purification Setup:**

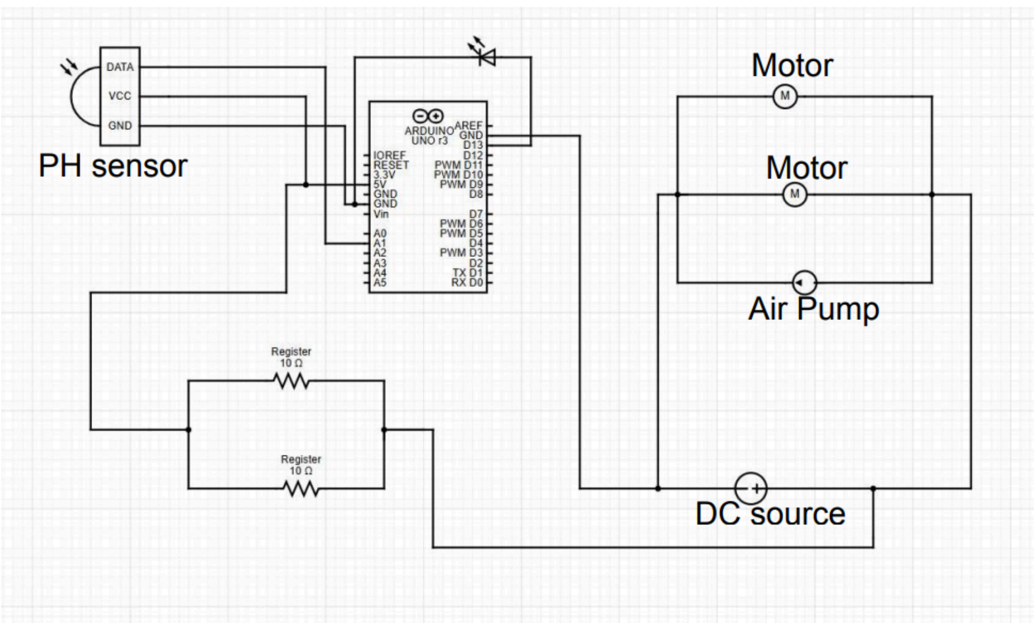
Develop the absorption unit within the system that uses sodium hydroxide as the primary reagent to capture and convert CO_2 from the atmosphere.

Optimize the chemical reaction efficiency by controlling factors like NaOH concentration, airflow rate, and temperature to maximize CO_2 capture.

Design a mechanism for easy replacement of sodium hydroxide when it becomes saturated with CO_2 and fully converts to sodium carbonate, ensuring continuous operation.

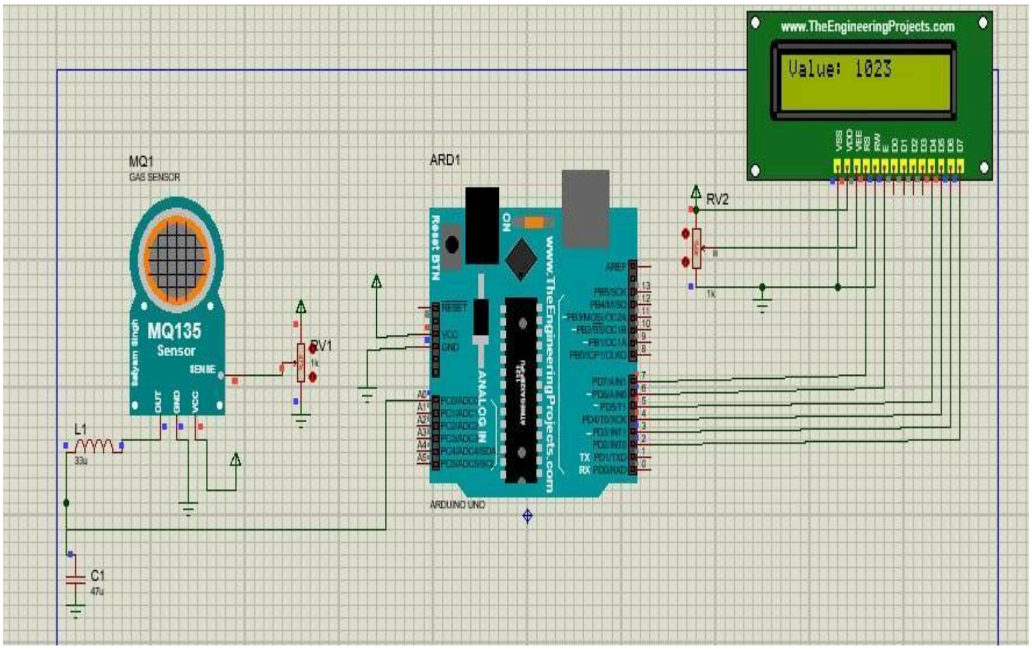
2. **Energy Supply Installation:**

Connect energy sources provide continuous operation for the system, ensuring an uninterrupted energy source



3. **Simulation using Proteus software**

- We have used Proteus 8.17 software to design a model circuit of the Co2 absorption system



4.2. Engineering Analysis

Engineering Analysis for the CO_2 Absorption System

This section presents preliminary engineering calculations for a CO_2 absorption system. The calculations focus on critical aspects, including airflow, power requirements, NaOH solution capacity, and CO_2 absorption efficiency. These analyses help ensure the system design is technically feasible and meets the required performance goals for capturing and processing CO_2.

1. Airflow Calculations

The airflow rate is crucial for efficient CO_2 absorption. It can be calculated using:

$Q = A * v$

Where:

- Q = Airflow rate (m^3/s)
- A = Cross-sectional area of the airflow path (m^2)
- v = Velocity of air (m/s)

Example Calculation:

- Diameter of the PVC pipe: 2.5 inches = 0.0635 m
- The radius of the pipe: $r = 0.03175$ m
- Area $A = \pi * r^2 = \pi * (0.03175)^2 \approx 0.00317\ m^2$
- Air velocity $v = 1.5$ m/s

Airflow rate:

$$Q = 0.00317 * 1.5 \approx 0.00475\ m^3/s$$

To convert to liters per minute (L/min):

$Q = 0.00475 * 60000 = 285$ L/min

This is the required airflow rate for the system.

2. Material and Structural Calculations

Next, we estimate the volume of the PVC chamber:

$V = L * A$

Where:

- L = Length of the PVC chamber (m)
- A = Cross-sectional area of the chamber (m^2)

For a 12-inch-long PVC chamber ($L = 0.3048$ m) and the calculated area $A = 0.00317\ m^2$:

$$V = 0.3048 * 0.00317 \approx 0.00097\ m^3 \text{ (or 0.97 liters)}$$

This gives us the volume of the PVC chamber.

3. Power Requirements

To calculate the total power consumption:

$P_{total} = V * I$

Where:

- V = Voltage (12V for the system)

- I = Total current draw (measured or from datasheet specifications)

By measuring the current of each system component, you can determine the total current draw and calculate the power consumption.

4. NaOH Solution Capacity

We estimate the amount of CO_2 that the NaOH solution can absorb using:

CO_2 absorbed = VNaOH * CNaOH * E

Where:

- VNaOH = Volume of NaOH solution (liters)
- CNaOH = Concentration of NaOH (mol/L)
- E = Efficiency of absorption

Example Calculation:

For a 2L, 1 mol/L NaOH solution with an efficiency E = 0.9:

CO_2 absorbed = 2 * 1 * 0.9 = 1.8 mol

Convert moles of CO_2 to grams (1 mol CO_2 = 44 grams):

1.8 * 44 = 79.2 grams of CO_2

This is the amount of CO_2 that the NaOH solution can absorb.

5. Efficiency Analysis

To determine the system's efficiency, we compare the expected CO_2 absorption rate with the airflow rate. Assuming the CO_2 concentration in the air is 400 ppm (parts per million):

- The mass of CO_2 per cubic meter of air is approximately 0.49 g/m³ (using air density = 1.225 kg/m³ at 25°C and sea level).
- We can calculate the mass of CO_2 passing through the system per second: $mCO_2 = Q * \text{Mass } CO_2 = 0.00475 * 0.49 \approx 0.00233$ g/s
- To get the mass per minute: 0.00233 g/s * 60 = 0.14 g/min

This value represents how much CO_2 passes through the system per minute.

These calculations ensure that the CO_2 absorption system is technically feasible, efficient, and able to meet the desired performance goals. By confirming the system's airflow, power needs, material strength, and CO_2 absorption capacity, we validate its effectiveness for real-world applications.

4.3. Economic analysis

The economic analysis evaluates the financial feasibility and cost-effectiveness of the CO_2 absorption system, balancing technical and economic criteria. It examines capital investments, operational costs, and long-term benefits like reduced emissions and sustainability contributions. This analysis highlights cost optimization opportunities and supports scalability for broader environmental applications. Initial costs, including expenses for electrical equipment, structural materials, and the NaOH solution, are detailed to provide a clear understanding of the system's financial requirements.

No.	Name of component	specification	Quantity	The cost of each item in ETB	Total cost in ETB
1	2V,1A Power adapter	Powers the device	1	734	734
2	fan	Drives airflow through the device	2		652
3	Arduino UNO	Processes the information that came from the sensor	1	2204	2204
4	Aquarium air pump	Circulates the NaOH to increase CO2 absorption	1	2000	2000
5	Jumper wire	Connecting	9	11	101

		The circuit			
6	resistor	Limit the current	2	11	22
7	Breadboard	Prototyping, testing, and connecting electronic components.	1	391	391
8	Air tube	Directs airflow through the system	1	200	200
9	Gas Sensor	To track the amount of Co2 level being absorbed	1	11	11
10	Plastic sheet	To separate the solution	1	70	70
11	NaOH	Absorbs CO_2 from the air	1kg	400	400
12	PVC Box	housing the NaOH solution	1	599	599
13	Glue stick	Adhesive for paper and crafts.	1	99	99
				Grand Total	7483

Table 1:Cost of material

4.3.1. Operating Costs

The operating costs of the CO_2 absorption system include recurring expenses to maintain its functionality and efficiency. These costs cover electricity consumption, NaOH solution replacement, and routine maintenance.

1. Electricity Consumption

The system operates with a 12V, 1A power supply, which consumes 12W of power:

- Power = Voltage × Current = 12V × 1A = 12W
- Energy consumption per day = 12W × 24h = 288Wh = 0.288kWh
- Monthly energy consumption = 0.288kWh × 30 = 8.64kWh

Using the Ethiopian electricity tariff of 0.48 Birr per kWh for up to 50 kWh consumption, the monthly electricity cost is:

- Monthly electricity cost = 8.64kWh × 0.48 ETB/kWh = 4.15 ETB/month

2. NaOH Replacement

The system uses 2 liters of NaOH solution, with replacement required every two weeks. The monthly NaOH cost is:

- Monthly NaOH cost = 2 liters × 2 replacements × 400 ETB/liter = 1600 ETB/month

3. Maintenance Costs

Routine maintenance, including cleaning components and checking system functionality, costs approximately 200 ETB per month.

Total Monthly Operating Costs

The total monthly operating costs for the system are:

- Total monthly operating cost = Electricity cost + NaOH replacement cost + Maintenance cost
- Total = 4.15 ETB + 1600 ETB + 200 ETB = **1804.15 ETB/month**

4.3.2. Cost-Benefit Analysis

This CO_2 absorption system is designed to efficiently capture CO_2 and provide a modular, scalable solution for localized carbon reduction. With the system's low initial and operating costs, it is a practical option for addressing environmental challenges, particularly in urban settings.

The system can be scaled for larger applications, and with long-term improvements, such as bulk manufacturing and material innovation, the performance and affordability can be further optimized to meet global sustainability goals.

CHAPTER FIVE

5. TESTING AND RESULT DISCUSSION

5.1. Testing

Multiple CO_2 concentration measurements were taken at both the inlet and outlet to assess the system's efficiency. Average values for both inlet and outlet CO_2 were calculated, and efficiency was determined based on the difference.

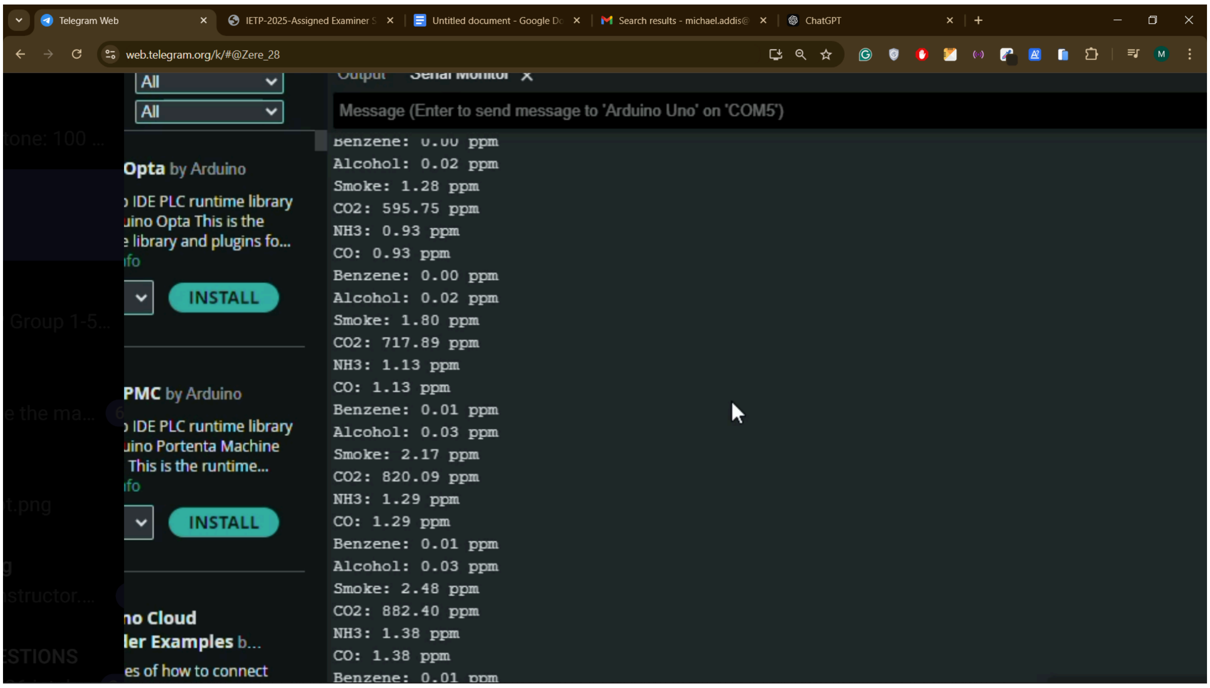


Figure 5.1 Co2 level in ppm from the test

Test	Inlet Co2(ppm)	OutletCo2(ppm)
1	456	200
2	500	310
3	480	370
4	473	363
5	700	320

Average CO_2 Concentrations:

- **Inlet CO_2**: 520 ppm
- **Outlet CO_2**: 272 ppm

Efficiency Calculation:

Efficiency = (520 - 272)/520 * 100

The system demonstrated an average efficiency of 47.69% in CO_2 absorption.

5.2. Analysis

The system demonstrated an average efficiency of 47.69% in CO_2 absorption. This suggests that the CO_2 removal process is moderately effective. However, a few factors could have influenced the results:

- **Air Dynamics**: The airflow through the system could have impacted the efficiency. Variations in air velocity or turbulence within the system might have affected the CO_2 absorption rate.
- **Sensor Calibration**: The accuracy of the CO_2 sensors is another potential factor. Any sensor drift or calibration issues could result in slight measurement errors.
- **System Operation**: The performance may also depend on the continuous operation of the fan. If the fan's performance fluctuated during the tests, it could have affected the airflow and, in turn, the absorption efficiency.

These factors should be considered in future tests to optimize the system's performance and enhance CO_2 absorption efficiency.

5.3. Result discussion

5.3.1. Challenges and Solutions

In any complex project, there are often technical and design challenges that arise, and our project was no exception. Some of the most prominent hurdles we faced were budget constraints, time limitations, and technical difficulties that required innovative solutions.

Budget Constraints

One of the primary challenges was managing within a limited budget. This often meant we had to make difficult decisions about which features to prioritize or adjust our approach to optimize costs. For instance, acquiring high-end materials or specialized equipment could have significantly increased our efficiency, but due to financial limitations, we had to source more cost-effective alternatives without compromising the overall functionality. the **Solution** that we came to minimize these problems was that we focused on maximizing the value of the resources we had. We strategically selected materials and tools that offered a balance between performance and affordability. Additionally, we leveraged existing technologies where possible and adopted open-source solutions to minimize software costs. We also sought partnerships and sponsorships that could alleviate some of the financial burdens, making it possible to stay on track with key milestones.

Time Constraints

Another critical challenge was the tight timeline. With limited time to complete the project, there was considerable pressure to execute tasks quickly without sacrificing quality. This led to a need for rapid decision-making, often requiring us to choose between the perfect solution and a practical one that could be implemented quickly. Approach Taken To manage the time crunch, we adopted an agile approach to development, breaking the project into smaller, manageable phases. This allowed us to continuously assess our progress and make real-time adjustments. We also held weekly stand-up meetings to ensure that team members were aligned and could address any roadblocks promptly. Prioritizing tasks based on impact and urgency also helped us stay focused on what needed to be done immediately versus what could be delayed.

Technical Difficulties

The technical challenges encountered in the project primarily revolved around system integration, the efficiency of CO_2 absorption, and the overall performance of the filtration system under real-world conditions. Issues such as inconsistent airflow, unexpected fluctuations in CO_2 capture efficiency, and occasional blockages in the aquarium pump hindered progress. Additionally, problems arose with the interaction between the NaOH solution and the fans, which affected the absorption rate and required troubleshooting to ensure consistent filtration performance.T o address these challenges, we employed a strategy of continuous testing and iterative improvements. First, we conducted thorough tests to evaluate the air intake efficiency and ensure that the fans were properly drawing air into the NaOH solution at an optimal rate. By adjusting fan speeds and nozzle placements, we were able to maximize air exposure to the solution. Regular monitoring and diagnostics of the aquarium pump allowed us to detect and address any blockages or pressure issues that could disrupt the mixing process. We also integrated sensors to measure CO_2 concentration in real-time, which helped us adjust fan and pump operation to maintain consistent air mixing and CO_2 absorption rate.. This modularity allowed us to quickly identify and resolve issues in specific components, ensuring smoother system operation and greater reliability in CO_2 filtration.

5.3.2. Environmental Impact and Practical Considerations

From an environmental standpoint, the CO_2 filtration system demonstrated the potential to contribute to CO_2 reduction goals. The ability to capture CO_2 from the air, especially in enclosed spaces or areas with high levels of CO_2 emissions, holds promise for reducing the carbon footprint. However, for the system to be widely adopted, practical considerations must be addressed. These include improving the system's scalability to handle larger volumes of air, reducing operational and maintenance costs, and ensuring that it can operate efficiently under varying environmental conditions.

The use of NaOH as the absorbent is effective in reducing CO_2, but it is important to note that NaOH is not a sustainable long-term solution without careful management. For instance, the disposal of the used NaOH solution could pose environmental concerns if not handled correctly. Additionally, finding a way to regenerate the NaOH solution for continuous use could significantly improve the system's sustainability.

5.3.3. Future Improvements

The results of the system's performance suggest several potential areas for improvement. First, enhancing the efficiency of air circulation and the mixing process will be critical for optimizing the CO_2 absorption rate. This could be achieved by using more advanced fan and pump systems that provide more consistent airflow and better mixing capabilities. Additionally, integrating sensors and automation systems that can adjust fan speed and pump operation based on real-time CO_2 levels could improve system responsiveness and overall performance.

Another area for improvement is the NaOH solution itself. Research into more sustainable or regenerative solutions could help address the issue of solution degradation over time, reducing the frequency of solution replacement. Exploring alternative absorbents that are more cost-effective and have longer lifespans could also help lower operational costs.

Finally, improving the scalability of the system to handle larger air volumes and making it more energy-efficient will be crucial for its practical deployment in real-world applications. As the system evolves, the goal will be to create a solution that can be easily integrated into different environments, whether in industrial settings or smaller, more localized applications.

CONCLUSION

The CO_2 capture and removal system demonstrated an average efficiency of 47.69% in filtering CO_2 from the air. While the system showed moderate effectiveness, there are opportunities for improvement. Variations in air dynamics, sensor accuracy, and fan performance could have influenced the results. To enhance the system's efficiency, future efforts should focus on optimizing airflow, ensuring precise sensor calibration, and maintaining consistent operation of the fan. Additional testing and adjustments will be crucial to refine the design and maximize CO_2 absorption, leading to a more effective air purification system.

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